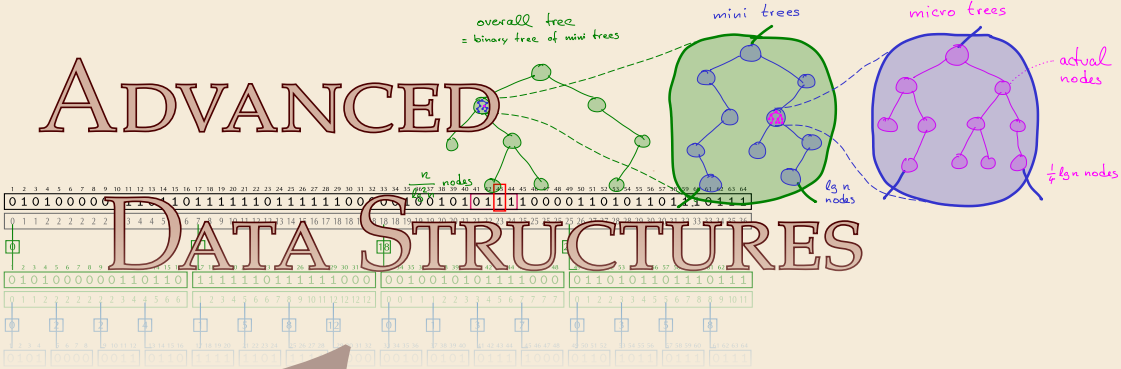
Advanced Data Structures



1 Randomized Trees

Advanced Data Structures · Summer 2026

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Outline

1 Randomized Trees

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1.2 What's wrong with BBSTs?
1.3 Random BSTs
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1.5 Treaps
1.6 Updates in Treaps
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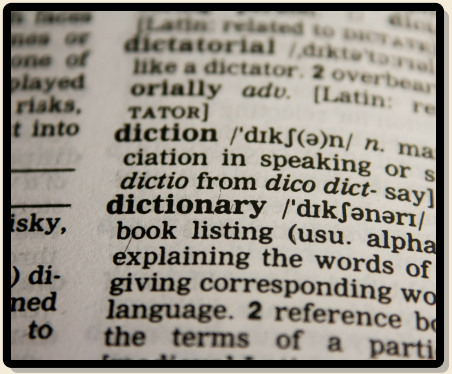
1.1 Recap: Sorted Dictionary and BSTs

Symbol table ADT

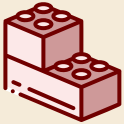
Java: `java.util.Map<K,V>`

Symbol table / Dictionary / Map / Associative array / key-value store:

Python dict `{k:v}`



- `put(`k`,`v`)` *Python* dict: `d[`k`] = `v
  Put key-value pair (k, v) into table
- `get(`k`)` *Python* dict: `d[`k`]`
  Return value associated with key k
- `delete(`k`)` *Python* dict: `del d[`k`]`
  Remove key k (any associated value) form table
- `contains(`k`)` *Python* dict: k `in d`
  Returns whether the table has a value for key k
- `isEmpty(), size()`
- `create()`

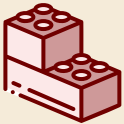


Most fundamental building block in computer science.

(Every programming library has a symbol table implementation.)

Ordered symbol tables

- `min()`, `max()`
  Return the smallest resp. largest key in the ST
- `floor(`x`)`, $\lfloor x \rfloor = \mathbb{Z}.$`floor(`$x$`)`
  Return largest key k in ST with $k \leq x$.
- `ceiling(`x`)`
  Return smallest key k in ST with $k \geq x$.
- `rank(`x`)`
  Return the number of keys k in ST $k < x$.
- `select(`i`)`
  Return the ith smallest key in ST (zero-based, i. e., $i \in [0..n)$)



With select, we can simulate access as in a truly dynamic array!

(Might not need any keys at all then!)

Binary search trees

Binary search trees (BSTs) $\approx$ dynamic sorted array

- binary tree
  - Each node has left and right child
  - Either can be empty (`null`)
- Keys satisfy ***search-tree property***

$$\boxed{\text{all keys in left subtree} \leq \text{root key} \leq \text{all keys in right subtree}}$$

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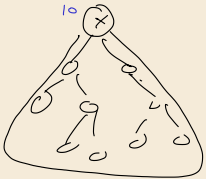
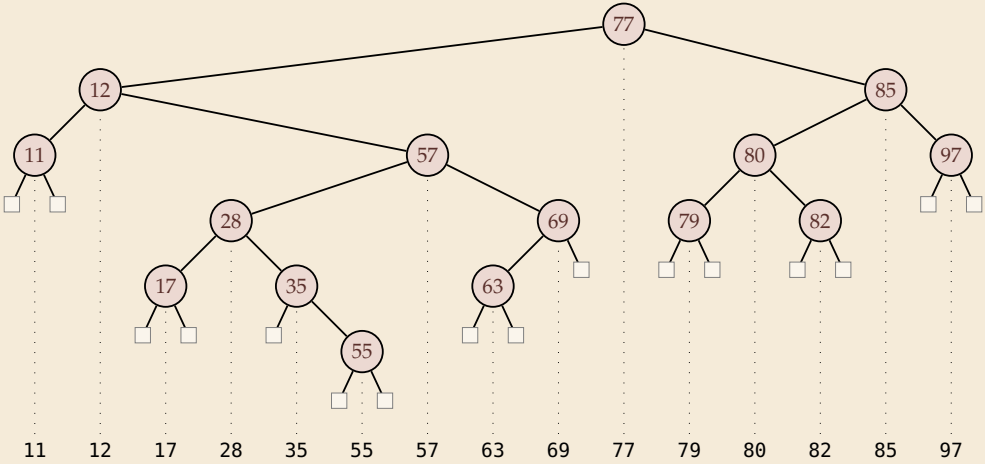
- Standard trick: ***Augmented BSTs***
  - Each node stores the **size** of its **subtree**
  - Easy to maintain upon updates
  - $\rightsquigarrow$ Allows to answer all ordered-symbol-table operations

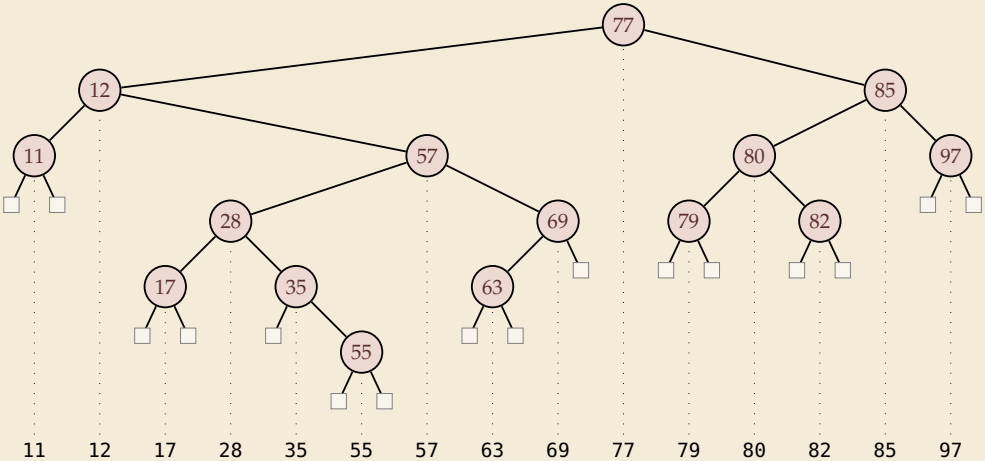


BST example & find



BST insert

Example: Insert 88



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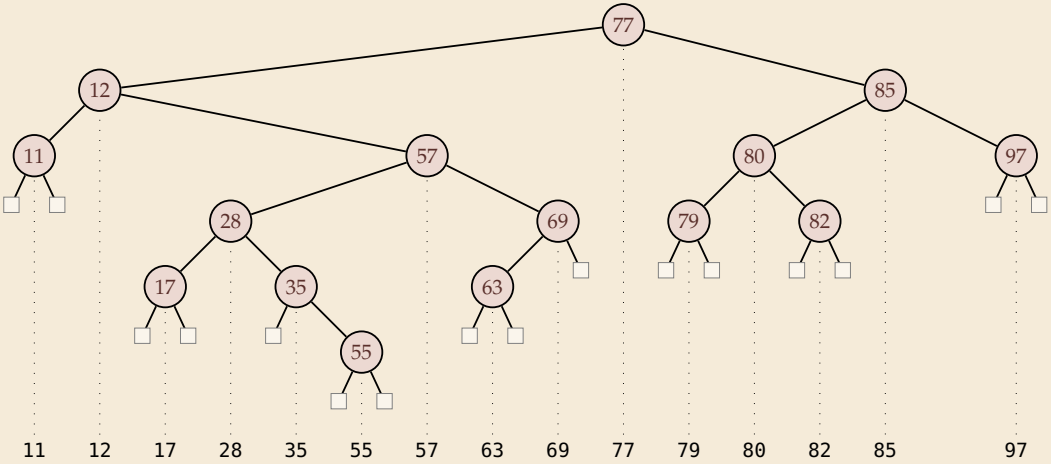

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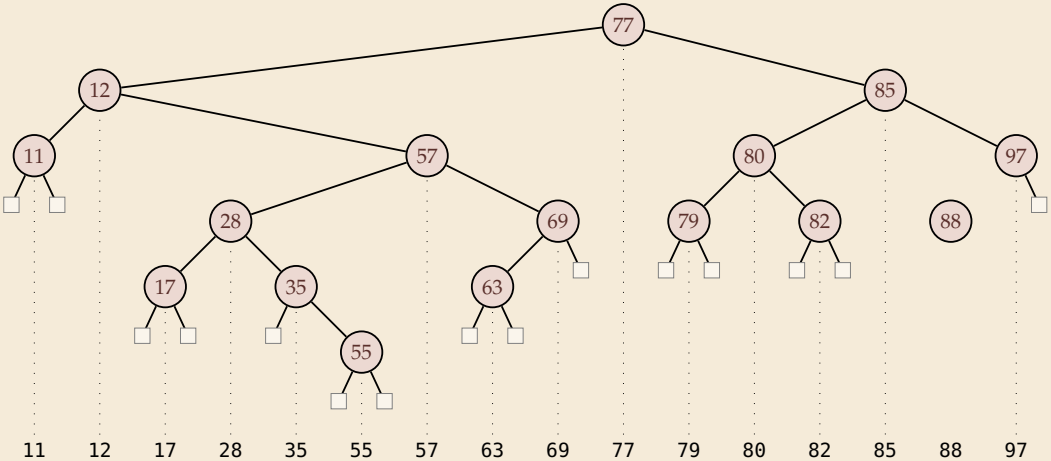

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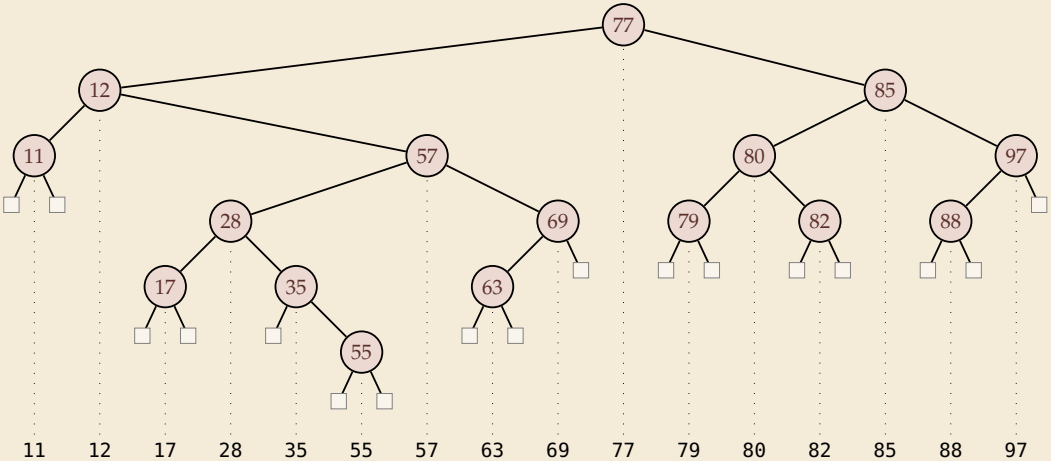
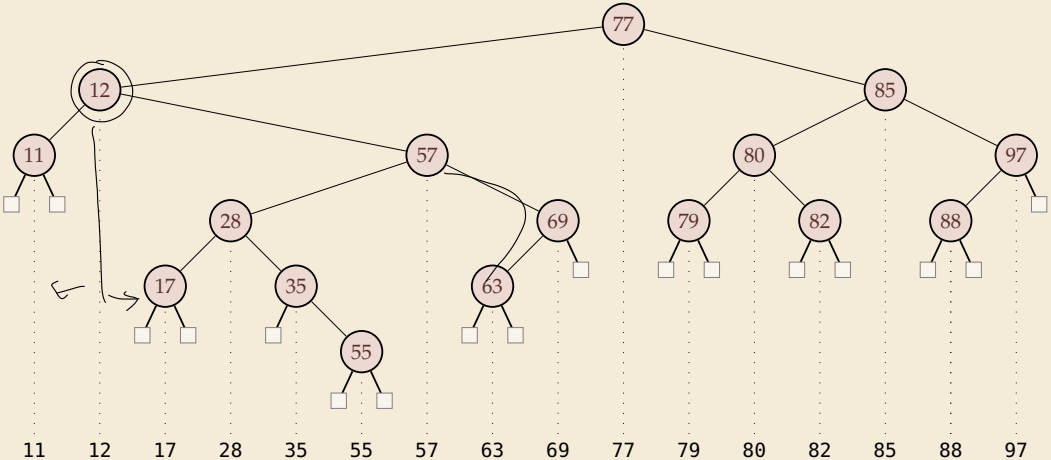


BST delete

Hibbard's deletion

- Easy case: remove leaf, e. g., 11 ⇝ replace by `null`
- Medium case: remove unary, e. g., 69 ⇝ replace by unique child
- Hard case: remove binary, e. g., 85 ⇝ swap with predecessor, recurse



Unbalanced Binary Search Trees

Operation	Running Time
`construct(`$A[1..n]$`)`	$O(n\boldsymbol{h})$
`put(`k`,`v`)`	$O(\boldsymbol{h})$
`get(`k`)`	$O(\boldsymbol{h})$
`delete(`k`)`	$O(\boldsymbol{h})$
`contains(`k`)`	$O(\boldsymbol{h})$
`isEmpty()`	$O(1)$
`size()`	$O(1)$
`min(),max()`	$O(1)$ (if stored)
`floor(`x`),ceiling(`x`)`	$O(\boldsymbol{h})$
`rank(`x`),select(`i`)`	$O(\boldsymbol{h})$

$h =$ *height* of the BST

$1 - 2^{-n}$

- Height h of unbalanced BST depends on insertion order
- satisfies $\lg n \leq h \leq n$

$\lg = \log_2$

$\text{Prob} = 1 - o(1)$

- For **random** insertions, $h = O(\log n)$ in expectation and with high probability

Balanced Binary Search Tree

Balanced binary search trees (BBSTs):

- imposes shape invariant that guarantees $O(\log n)$ height
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  - *red-black trees*

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  - *weight-balanced trees* (BB[α] trees)

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- **Properties**
  - All of them guarantee $h = O(\log n)$ at all times (constants differ!)
  - All of them work with *rotations* along the search path

  $\rightsquigarrow$ $O(\log n)$ extra cost for maintaining balance
  - Further guarantees specific to rule known

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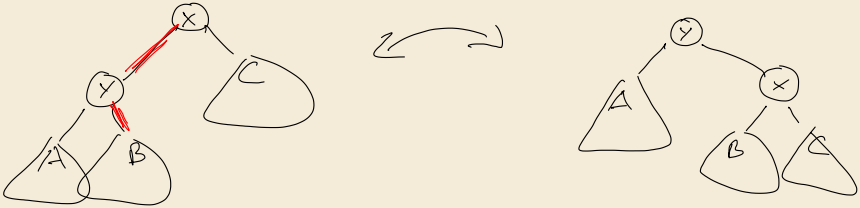
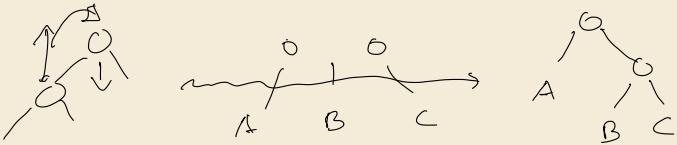
OPEN: *What is the exact average cost (constant in front of $\lg n$) of insertion in AVL trees / red-black trees?*

A Primitive for BSTs: Rotations





1.2 What's wrong with BBSTs?

Red-Black Tree Implementation

`RedBlackTree` from Sedgewick & Wayne (excerpt) `algs4.cs.princeton.edu/code`

- showing only the bare-bones core: `put` (insert) and `delete`
- no rank-based operations
- full class has 750 lines (and this is a good and compact implementation!)

```
public class RedBlackBST<Key extends Comparable<Key>, Value> {
    public void put(Key key, Value val) {
        root = put(root, key, val);
        root.color = BLACK;
    }

    private Node put(Node h, Key key, Value val) {
        if (h == null) return new Node(key, val, RED, 1);

        int cmp = key.compareTo(h.key);
        if      (cmp < 0) h.left  = put(h.left,  key, val);
        else if (cmp > 0) h.right = put(h.right, key, val);
        else              h.val   = val;

        // fix-up any right-leaning links
        if (isRed(h.right) && !isRed(h.left))      h = rotateLeft(h);
        if (isRed(h.left)  &&  isRed(h.left.left)) h = rotateRight(h);
        if (isRed(h.left)  &&  isRed(h.right))     flipColors(h);
        h.size = size(h.left) + size(h.right) + 1;

        return h;
    }

    private Node deleteMin(Node h) {
        if (h.left == null)
            return null;

        if (!isRed(h.left) && !isRed(h.left.left))
            h = moveRedLeft(h);

        h.left = deleteMin(h.left);
        return balance(h);
    }
```

```
    private Node deleteMax(Node h) {
        if (isRed(h.left))
            h = rotateRight(h);

        if (h.right == null)
            return null;

        if (!isRed(h.right) && !isRed(h.right.left))
            h = moveRedRight(h);

        h.right = deleteMax(h.right);

        return balance(h);
    }

    private Node delete(Node h, Key key) {
        // assert get(h, key) != null;

        if (key.compareTo(h.key) < 0)  {
            if (!isRed(h.left) && !isRed(h.left.left))
                h = moveRedLeft(h);
            h.left = delete(h.left, key);
        }
        else {
            if (isRed(h.left))
                h = rotateRight(h);
            if (key.compareTo(h.key) == 0 && (h.right == null))
                return null;
            if (!isRed(h.right) && !isRed(h.right.left))
                h = moveRedRight(h);
            if (key.compareTo(h.key) == 0) {
                Node x = min(h.right);
                h.key = x.key;
```

```
                h.val = x.val;
                // h.val = get(h.right, min(h.right).key);
                // h.key = min(h.right).key;
                h.right = deleteMin(h.right);
            }
            else h.right = delete(h.right, key);
        }
        return balance(h);
    }

    private Node rotateRight(Node h) {
        assert (h != null) && isRed(h.left);
        Node x = h.left;
        h.left = x.right;
        x.right = h;
        x.color = h.color;
        h.color = RED;
        x.size = h.size;
        h.size = size(h.left) + size(h.right) + 1;
        return x;
    }

    private Node rotateLeft(Node h) {
        assert (h != null) && isRed(h.right);
        Node x = h.right;
        h.right = x.left;
        x.left = h;
        x.color = h.color;
        h.color = RED;
        x.size = h.size;
        h.size = size(h.left) + size(h.right) + 1;
        return x;
    }
```

```
    private void flipColors(Node h) {
        h.color = !h.color;
        h.left.color = !h.left.color;
        h.right.color = !h.right.color;
    }

    private Node moveRedLeft(Node h) {
        flipColors(h);
        if (isRed(h.right.left)) {
            h.right = rotateRight(h.right);
            h = rotateLeft(h);
            flipColors(h);
        }
        return h;
    }

    private Node moveRedRight(Node h) {
        flipColors(h);
        if (isRed(h.left.left)) {
            h = rotateRight(h);
            flipColors(h);
        }
        return h;
    }

    private Node balance(Node h) {
        if (isRed(h.right) && !isRed(h.left))    h = rotateLeft(h);
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```

Can we have something simpler?

In this and the next unit, we will look into alternatives to avoid lengthy case distinctions.

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Disclaimer: It seems that something has to give.

All known (much) more elegant BST alternatives are either *randomized* or *amortized*.

- However, this is often tolerable.
- Complexity of code can also come with running-time penalty;
  if we avoid that, might be an overall speedup.
  (Typically not the case for well-tested library implementations, though.)

Recall: Average-Case Analysis vs. Randomized Algorithms

This unit: *Randomized solutions*.

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- algorithm is **deterministic**
  same input, same computation

Randomized Algorithm (here)

- algorithm is **not** deterministic
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Confusingly enough, the analysis (technique) is often the same!

But: Implications are quite different; randomization is much more versatile and robust.

1.3 Random BSTs

Random BSTs are good (enough)

*Let's first do some wishful thinking: assume we're dealing with **random** inputs only.*

- ⇝ Let's first see if we like the performance in this case.
- If so, will try and see how to **enforce** this randomness later.

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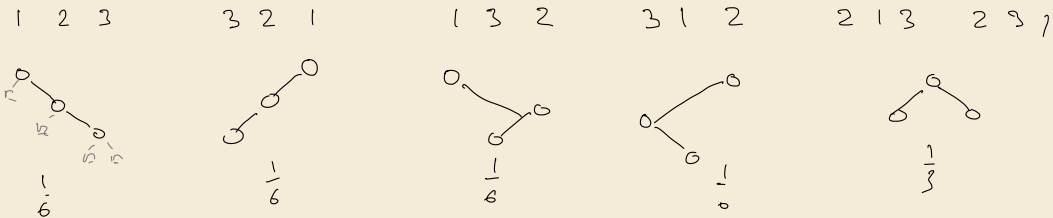
Random here means:

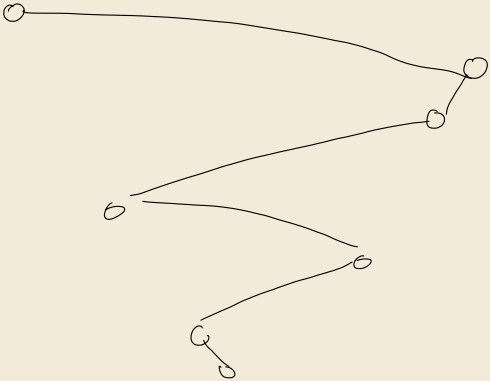
Definition 1.1 (Random BST)

A ***Random BST*** of size n is the (random) shape of an initially empty BST after successively inserting a random permutation of $[n]$.

$= \{1, \dots, n\}$

Example: $n = 3$





$\approx 2^n$ maximally unbalanced trees

#binary trees on n nodes

nth Catalan number

$$= \binom{2n}{n} \cdot \frac{1}{n+1}$$

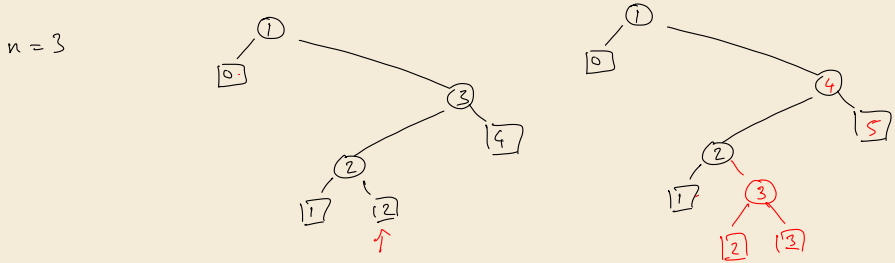
$$\approx 4^n / n^{3/2}$$

$$n! \approx \left(\frac{n}{e}\right)^n$$

Iterative randomness

Lemma 1.2 (Random insertion yields random BST)

Let $n \geq 0$ and T_n a random BST over n keys. Inserting an element equally likely in one of the $n+1$ *gaps* in T_n (external leaves) results in a new BST T_{n+1} that has the same shape as a random BST on $n+1$ keys.



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Proof:

Insertion order for T_{n+1} is a permutation of $[n+1] = \{1, \ldots, n+1\}$, $[0..n]$ which is in bijection with a permutation of $[n]$ and a *insertion point (leaf)* in $[0..n]$, where we increment all values right of the insertion leaf by 1.

$= [1..n+1]$

$= \{1, 2, \ldots, n+1\}$

Example: $1, 2, 4, 7, 6, 5 \mid 3 \;\widehat{=}\; ((1, 2, 3, 6, 5, 4), 2)$

Recall: Events

something we can assign a probability to

$A \in \mathcal{F}$ is called an *event* of probability space $(\Omega, \mathcal{F}, \mathbb{P})$; also a *measurable set*.

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Basic properties

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- $\mathbb{P}\left[\bigcup A_i\right] \leq \sum_i \mathbb{P}[A]$ the *union bound* (a.k.a. Boole's inequality a.k.a. σ-subadditivity)

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- $\{A_1, \ldots, A_k\}$ *(mutually) independent* $\iff \mathbb{P}\left[\bigcap_i A_i\right] = \prod_i \mathbb{P}[A_i]$

  An infinite set of events is mutually independent if every finite subset is so.

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discrete, $\mathcal{F} = 2^{\Omega}$ (powerset of Ω)

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  An infinite set of events is mutually independent if every finite subset is so.

  k-wise independence means that only all size-k subsets are independent.
- *conditional probability* for A given B: $\mathbb{P}[A \mid B] = \mathbb{P}[A \cap B] / \mathbb{P}[B]$
  generally undefined if $\mathbb{P}[B] = 0$
- *law of total probability*: If $\Omega = B_1 \dot\cup B_2 \dot\cup \cdots$ is a partition of Ω, we have

$$\mathbb{P}[A] = \sum_{\substack{i \\ \mathbb{P}[B_i] \neq 0}} \mathbb{P}[A \mid B_i] \cdot \mathbb{P}[B_i].$$

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Random variables (r.v.) $X : \Omega \to \mathcal{X}$; often $\mathcal{X} = \mathbb{R}$ (in general spaces: only *measurable* functions)

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Basic properties and conventions:

- event $\{X = x\}$ is defined as $\{\omega \in \Omega : X(\omega) = x\}$.
- For event A define the indicator r.v. $\mathbb{1}_A$ via $\mathbb{1}_A(\omega) = [\omega \in A] = \begin{cases} 1 & \omega \in A \\ 0 & \text{else} \end{cases}$

Recall: Random Variables

Random variables (r.v.) $X : \Omega \to \mathcal{X}$; often $\mathcal{X} = \mathbb{R}$ (in general spaces: only *measurable* functions)

Basic properties and conventions:

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Independence:

- X and Y independent $\iff \mathbb{P}[X = x \wedge Y = y] = \mathbb{P}[X = x] \cdot \mathbb{P}[Y = y]$ for all x, y.

  (Naturally follows from independent events)

- a sequence of *i.i.d.* r.v. $X_1, X_2, \ldots$ (*independent and identically distributed*) has $X_i \overset{\mathcal{D}}{=} X_1$ and $\{X_i\}_{i \geq 1}$ are mutually independent
  - typical example: sequence of coin tosses (with same coin)

Alternative Random Model

Corollary 1.3

A BST built by inserting n i.i.d. Uniform$(0, 1)$ r.v. has the shape of a random BST.

1.4 Properties of Random BSTs

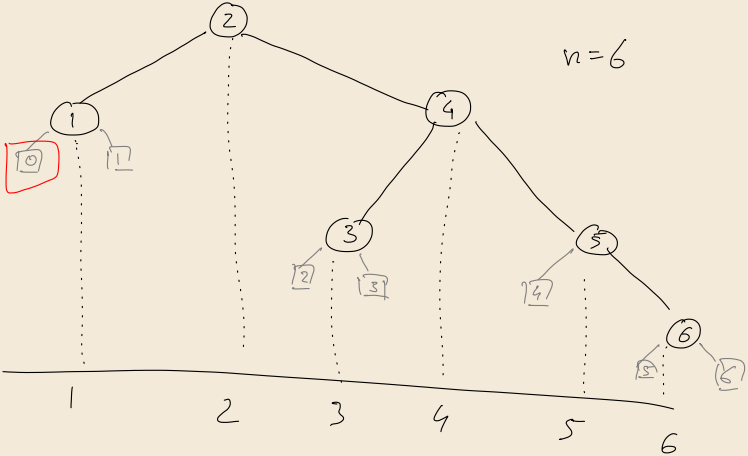


Figure 6.16 A binary search tree built from 237 randomly ordered keys

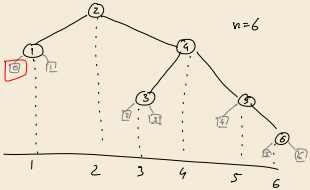
Analysis of Random BSTs I

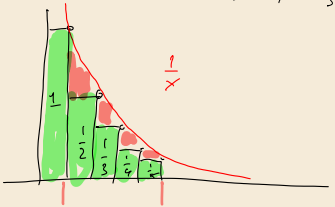
$$\frac{H_n}{\ln n} \to 1 \quad (n \to \infty)$$

Theorem 1.4 (Expected depth of leftmost leaf)

The *expected depth* (number of edges from root) of the leftmost external leaf (leaf for $-\infty$) in a random BST on $n \geq 1$ nodes is $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \sim \ln n$.

$$= \sum_{i=1}^{n} \frac{1}{i}$$





$$\int_1^n \frac{1}{x}\,dx \approx \ln(n) - \ln(1)$$

Analysis of Random BSTs I

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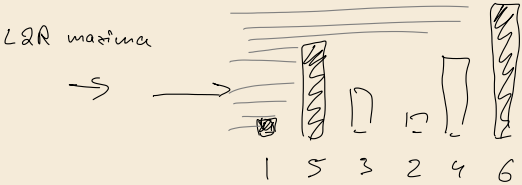
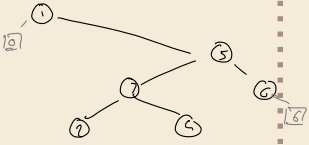
Proof:

$\text{depth}(\boxed{0}) = \#\text{left-to-right minima } \text{L2R}_n \text{ in insertion sequence}$

$\text{L2R}_n = \sum_{i=1}^{n} X_i \qquad \text{for } X_i = [\text{position } i \text{ is a l2r min}]$





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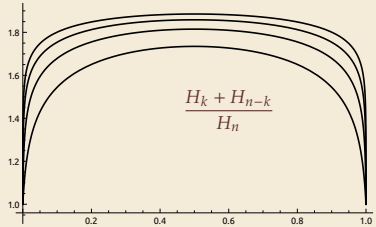
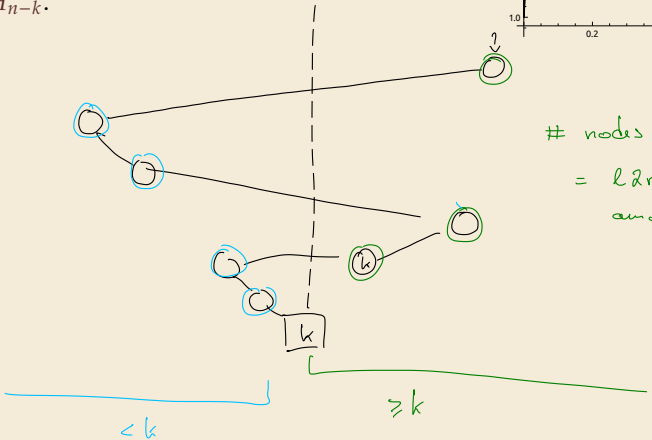
$$\mathbb{E}[\text{L2R}_n] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} \frac{1}{i} = H_n.$$ ■

Analysis of Random BSTs II

Theorem 1.5 (Expected depth of kth leaf)

The *expected depth* of the *kth external leaf* $\boxed{k}$ (for $k = 0, \ldots, n$) in a random BST on $n \geq 1$ keys is $H_k + H_{n-k}$.

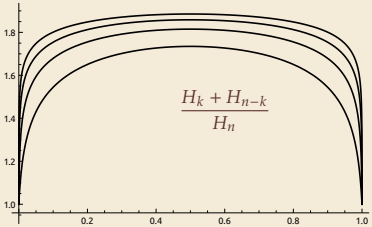




nodes right of k

= l2r minima among keys ≥ k

Analysis of Random BSTs II



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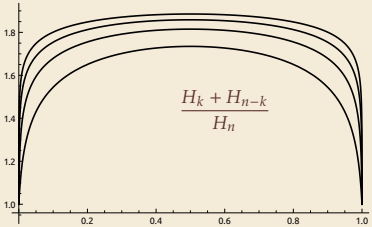
Analysis of Random BSTs II



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Proof:

$k - \frac{1}{2}$

$$\begin{aligned}
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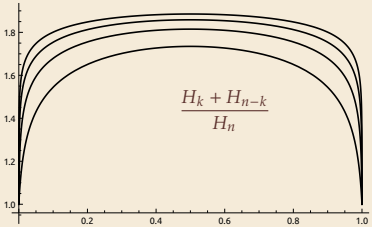
Analysis of Random BSTs II



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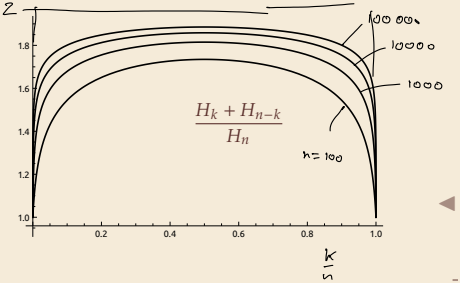

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&= \text{l2r-mins among keys} < x \;+\; \text{l2r-maxs with keys} > x
\end{aligned}$$

Since there are k keys $< x$ and $n - k$ keys $> x$, the same derivation as above applies. ■

Analysis of Random BSTs III

$$\mathbb{E}\left[\sum_{k=0}^{n} \text{depth}(\boxed{k})\right] = \sum_{k=0}^{n} (H_k + H_{n-k})$$

$$= 2\sum_{k=0}^{n} H_k$$

Corollary 1.6 (Depth of typical leaf)

Consider a random BST T_n of n keys.

$$\sum_{i=1}^{n} H_i = (n+1)H_n - n,$$

(a) The *expected external path length* of T_n is
$2(n+1)(H_{n+1} - 1) \;=\; 2n \ln n - 2(1-\gamma)n \;\pm\; O(\log n)$. ($\gamma \approx 0.5772$ the *Euler-Mascheroni constant*)

(b) The depth of the αnth leaf in a random BST of n keys $\sim 2 \ln n$ as $n \to \infty$ for any fixed $\alpha \in (0,1)$.

$$H_{n+1} = H_n + \frac{1}{n+1}$$

recursion tree of Quicksort

$$H_k + H_{n-k} = H_{\alpha n} + H_{(1-\alpha)n}$$

$$\sim \ln(\alpha n) + \ln((1-\alpha)n)$$

$$= 2\ln(n) + \ln(\alpha) + \ln(1-\alpha)$$

$$\sim 2\ln(n)$$

Detour: When Expectation Isn't Enough

- Two hypothetical algorithms:
  - A takes 1 step in half the cases and 3 steps otherwise
  - B takes 1 step in 99% of cases and **101** steps otherwise
  - $\rightsquigarrow$ both have expected costs of 2 steps.
  - probably want A ... certainly would want to be able to distinguish them!

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  - probably want A ... certainly would want to be able to distinguish them!

- **Goal:** Strengthen algorithms so $time(x)$ rarely far from $\mathbb{E}[time(x)]$
  - formally: bound probability that X (far) exceeds $\mathbb{E}[X]$
    - $\rightsquigarrow$ *concentration bounds* a.k.a. *tail inequalities*
  - 👍 can then compare these typical times again
  - 👍 also obtain more reliable algorithms
  - $\rightsquigarrow$ Let's establish some tools for that!

Detour: With High Probability

Definition 1.7 (With high probability)

We say

- an event $X = X(n)$ happens *with high probability (w.h.p.)* when $\forall c \;:\; \mathbb{P}[X(n)] \doteq 1 \pm O(n^{-c})$ as $n \to \infty$. $1 - d(c) \cdot n^{-c}$
- a random variable $X = X(n)$ is *in $O(f(n))$ with high probability (w.h.p.)* when $\forall c \; \exists d \;:\; \mathbb{P}[X \leq d f(n)] = 1 \pm O(n^{-c})$ as $n \to \infty$.
  (This means, the constant in $O(f(n))$ may depend on c.)

Height / Depth of a leaf in Random BST $H = O(\log n)$ w.h.p.

$c = 10 \rightsquigarrow d_1, d_2 : \mathbb{P}[H \geq d_1 \lg n] \leq d_2 n^{-10}$

(for example, $\mathbb{P}[\text{height} \geq 42 \ln n] \leq 2 n^{-7.4}$)

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  (This means, the constant in $O(f(n))$ may depend on c.) ◀
- Very strong notion: failure probability smaller than any polynomial

$\rightsquigarrow$ If A succeeds w.h.p. then also polynomially many repetitions of A succeed w.h.p.

Lemma 1.8 (Repetitions w.h.p.)

Suppose A is an algorithm that w.h.p. does not fail.
In n^d *independent* repetitions of A on inputs of size n, w.h.p. *no* repetition fails. ◀

Detour: With High Probability [2]

Proof (Lemma 1.8):

Let c from the definition of w.h.p. be given.

$\rightsquigarrow$ events that happen with high probability can be combined

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Detour: Chernoff Bounds

Strong concentration inequalities require assumptions on distribution of X.
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Theorem 1.9 (Chernoff Bound for Bernoulli trials)

Let $X_1, \ldots, X_n \in \{0,1\}$ be *(mutually) independent* with $X_i \stackrel{\mathcal{D}}{=} \mathrm{B}(p_i)$.
Define $X = X_1 + \cdots + X_n$ and $\mu = \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n] = p_1 + \cdots + p_n$. Then holds

$$\forall \delta > 0 \quad : \quad \mathbb{P}[X \geq (1+\delta)\mu] \;<\; \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}$$

$$\forall \delta \in (0,1] \quad : \quad \mathbb{P}[X \geq (1+\delta)\mu] \;\leq\; \underline{\exp(-\mu\delta^2/3)}$$

Proof (Sketch):

Apply Markov's Inequality to $Y = e^{tX}$, then choose convenient t.

(more details in *Advanced Algorithms*)

Detour: Chernoff Bound for Binomial Distribution

The algorithmically most widely used special case has identical coin flips.

Corollary 1.10 (Chernoff Bound for Binomial Distribution)

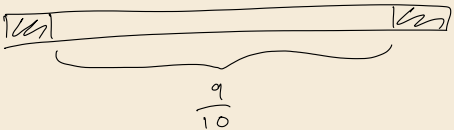
Let $X \stackrel{\mathcal{D}}{=} \mathrm{Bin}(n,p)$. Then

$$\forall \delta \geq 0 \quad : \quad \mathbb{P}\left[\left|\frac{X}{n} - p\right| \geq \delta\right] \leq 2\exp(-\overbrace{2\delta^2 n}^{\to\infty})$$

$$\delta > \frac{1}{\sqrt{n}} = n^{-0.5} \qquad n^{-0.499}.$$

$\rightsquigarrow$ $\boxed{\mathrm{Bin}(n,p) \in np \pm n^{0.501} \text{ w.h.p.}}$

$p = 0.1$



Analysis of Random BSTs – Concentration

Theorem 1.11 (Concentration of left-to-right minima)

The number of left-to-right minima in a permutation of length n is in $O(\log n)$ w.h.p.
Hence, the above expected results hold with high probability (up to constant factors).

Proof (Idea):
Via *inversion table* of permutation:
$a_1, \ldots, a_n$ permutation of $[n]$ in bijection with $b_1, \ldots, b_n$ where

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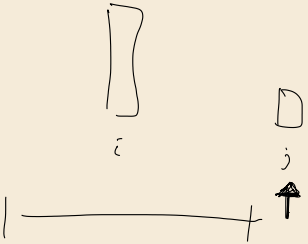

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*greater and left of *number* j*

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$$\text{L2RMax}(a) = \sum_{j=1}^{n} [b_j = 0] \rightsquigarrow \text{a sum of independent Bernoulli trials.}$$

+ symmetry

Hook-Length Formula for Random BSTs

For a given tree, the probability to see this shape can be computed recursively over the tree structure:

Theorem 1.12 (Random BST Distribution)

Let T_n be a binary tree. The probability that Random BSTs attains the shape T_n after n insertions is

$$\Pr[T_n] = \begin{cases} 1 & n = 0, \\ \frac{1}{n} \cdot \Pr[T_L] \cdot \Pr[T_R] & n \geq 1, \end{cases}$$ (for T_L and T_R the left resp. right subtrees of T).

randomness preservation! filtering out values $\leq x$

again a random permutation

$\mathbb{P}\left[\begin{array{c} \circ \\ \circ \quad \circ \end{array}\right] = \frac{1}{3}$ $\qquad \frac{1}{3}$

$\mathbb{P}\left[\circ \to \circ \to \circ\right] = \frac{1}{3} \cdot \mathbb{P}\left[\circ \to \circ\right] = \frac{1}{3} \cdot \frac{1}{2} \cdot 1 = \frac{1}{6}$

More AofA (Analysis of Algorithms)

Remark 1.13 (Knowledge on Random BSTs)

Random BSTs are extremely well-studied in Analysis of Algorithms.
A few more results (all proven in the literature):

(a) The **expected height** is $\alpha \ln n - \beta \ln \ln n \pm O(1)$ with $\alpha \approx 4.311$ and $\beta \approx 1.953$.

(b) The **height** divided by $\ln n$ **converges in probability** to the constant α.

(c) The number X_{nk} of **external leaves at depth** k satisfies $\mathbb{E}[X_{nk}] = \frac{2^k}{n!}\left[{n \atop k}\right]$.

(d) The **depth** of a typical **leaf** divided by $\ln n$ **converges in probability** to 2.

(e) The standardized **depth** of a random leaf **converges** in distribution to a standard **normal distribution**.

(f) The same is true for the standardized depth of a random internal node.

(g) Let D_n be the **depth of the nth inserted node**. Then $(D_n - \ln n)/\sqrt{\ln n}$ converges in distribution to a standard **normal distribution**. ◀

⇝ In many ways, random BSTs are close to perfectly balanced.

⇝ If only we can get the performance of random BSTs, we're happy.